Lab-6: JavaScript: use of document.writeln(), comments, variables, local vs Global, type of operators, arithmetic, assignment, relational, logical & Bitwise Operators, Control structures, conditional statements, array, user define function, object, use if For/in .

Code:

```
<!DOCTYPE html>
<html>
<body>

<script>

document.write("<b>Exercise 1: Operators</b><br>");

var a = 2, b = 3;
document.write("2 + 3 = " + (a + b) + "<br>");
document.write('"hello" + " bvm" = ' + ("hello" + " bvm") + "<br>");
document.write('2 + "MB" = ' + (2 + "MB") + "<br>");
document.write("3 - 2 = " + (b - a) + "<br>");
document.write('3 - "MB" = ' + (b - "MB") + "<br>");
document.write("3 * 2 = " + (b * a) + "<br>");
document.write("5 / 2 = " + (5 / 2) + "<br>");
```

```
document.write("2 / 0 = " + (2 / 0) + "<br>");
document.write("5 % 2 = " + (5 % 2) + "<br>");
document.write("2 % 0 = " + (2 % 0) + "<br>");

var x = 1;
document.write("x = 1; x++ = " + (x++) + "<br>");
document.write("x now = " + x + "<br>");
var y = ++x;
document.write("y = ++x; y=" + y + " x=" + x + "<br>");
var y2 = x++;
document.write("y = x++; y=" + y2 + " x=" + x + "<br>");
x = 5;
document.write("x = 5; x-- = " + (x--) + "<br>");
document.write("x now = " + x + "<br>");
var y3 = --x;
document.write("y = --x; y=" + y3 + " x=" + x + "<br>");

document.write("<br><b>Exercise 2: Bitwise Operators</b><br>");

var a = 5, b = 3;
document.write("a = 5, b = 3<br>");
document.write("a & b (AND) = " + (a & b) + "<br>");
document.write("a | b (OR)  = " + (a | b) + "<br>");
```

```
document.write("a ^ b (XOR) = " + (a ^ b) + "<br>");
document.write("~a (NOT)   = " + (~a) + "<br>");
document.write("a << 1 (Left Shift)  = " + (a << 1) + "<br>");
document.write("a >> 1 (Right Shift) = " + (a >> 1) + "<br>");

</script>

<br>
<b>Exercise 3: Fibonacci Numbers</b><br>
Enter N: <input type="number" id="fibN">
<button onclick="showFib()">Show</button>
<p id="fibOut"></p>

<b>Exercise 4: Prime Numbers in a Range</b><br>
Start: <input type="number" id="primeStart">
End: <input type="number" id="primeEnd">
<button onclick="showPrime()">Show</button>
<p id="primeOut"></p>

<b>Exercise 5: Array Methods</b><br>
Enter numbers (comma separated): <input type="text" id="arrInput">
<button onclick="showArrayMethods()">Show</button>
<p id="arrOut"></p>
```

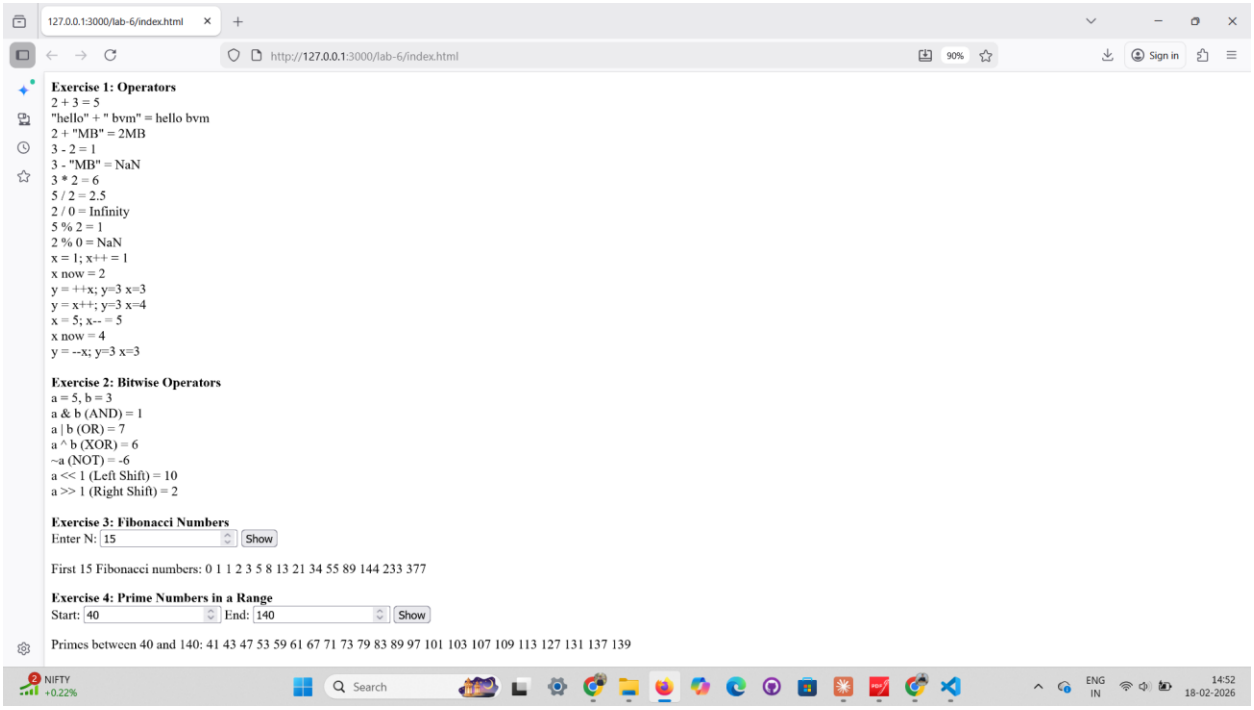
```
<b>Exercise 6: Sort Array</b><br>
Enter numbers (comma separated): <input type="text" id="sortInput">
<button onclick="showSort()">Show</button>
<p id="sortOut"></p>

<script>

function showFib() {
  var n = parseInt(document.getElementById("fibN").value);
  if (isNaN(n) || n <= 0) {
    document.getElementById("fibOut").innerHTML = "Please enter a valid
number.";
    return;
  }
  var f1 = 0, f2 = 1, temp;
  var result = "";
  for (var i = 0; i < n; i++) {
    result += f1 + " ";
    temp = f1 + f2;
    f1 = f2;
    f2 = temp;
  }
```

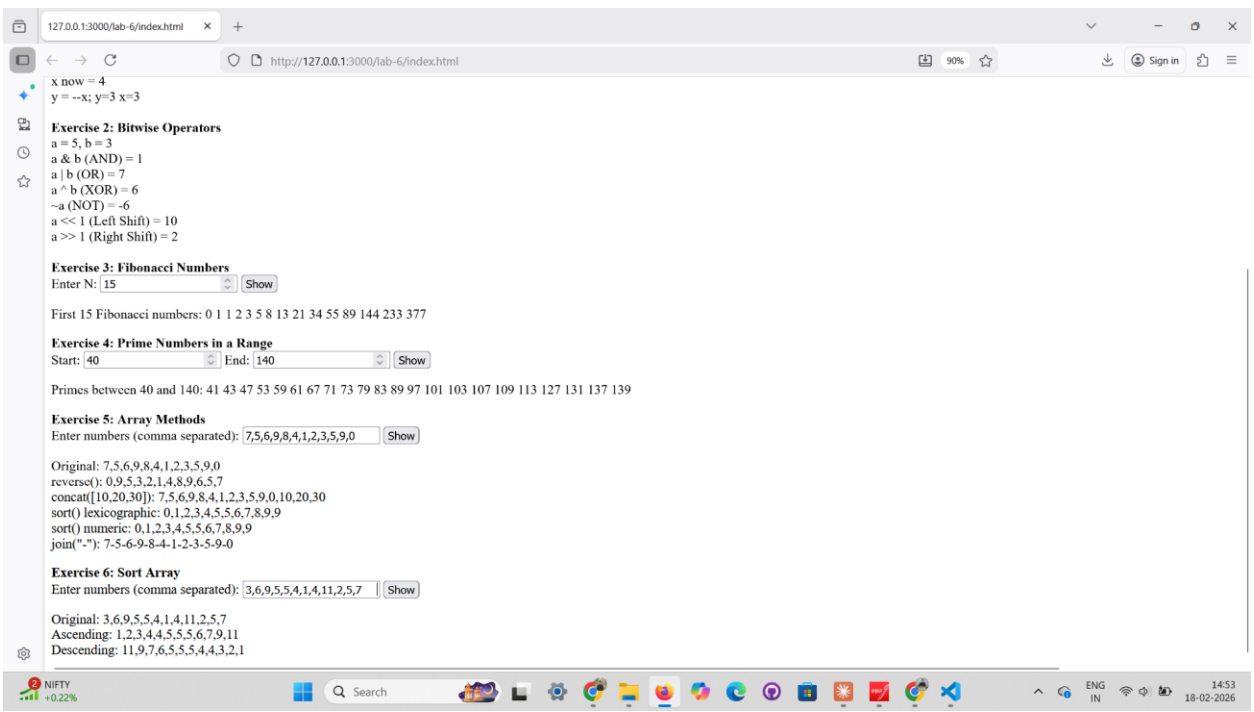
```
  document.getElementById("fibOut").innerHTML = "First " + n + " Fibonacci
numbers: " + result;
}


function showPrime() {
  var start = parseInt(document.getElementById("primeStart").value);
  var end = parseInt(document.getElementById("primeEnd").value);
  if (isNaN(start) || isNaN(end) || start > end) {
    document.getElementById("primeOut").innerHTML = "Please enter a valid
range.";
    return;
  }
  var result = "";
  for (var i = start; i <= end; i++) {
    var isPrime = true;
    if (i < 2) isPrime = false;
    for (var j = 2; j <= Math.sqrt(i); j++) {
      if (i % j == 0) { isPrime = false; break; }
    }
    if (isPrime) result += i + " ";
  }
  document.getElementById("primeOut").innerHTML = "Primes between " +
start + " and " + end + ": " + result;
}
```

```
function showArrayMethods() {
  var input = document.getElementById("arrInput").value;
  var arr = input.split(",").map(function(x){ return parseInt(x.trim()); });
  if (arr.some(isNaN)) {
    document.getElementById("arrOut").innerHTML = "Please enter valid
numbers.";
    return;
  }
  var out = "";
  out += "Original:  " + arr + "<br>";
  out += "reverse(): " + arr.slice().reverse() + "<br>";
  out += "concat([10,20,30]): " + arr.concat([10,20,30]) + "<br>";
  out += "sort() lexicographic: " + arr.slice().sort() + "<br>";
  out += "sort() numeric: " + arr.slice().sort(function(a,b){ return a - b; }) +
"<br>";
  out += 'join("-"): ' + arr.join("-") + "<br>";
  document.getElementById("arrOut").innerHTML = out;
}

function showSort() {
  var input = document.getElementById("sortInput").value;
  var arr = input.split(",").map(function(x){ return parseInt(x.trim()); });
  if (arr.some(isNaN)) {
```

```
   document.getElementById("sortOut").innerHTML = "Please enter valid
numbers.";
   return;
  }
  var out = "";
  out += "Original:   " + arr + "<br>";
  out += "Ascending:  " + arr.slice().sort(function(a,b){ return a - b; }) + "<br>";
  out += "Descending: " + arr.slice().sort(function(a,b){ return b - a; }) + "<br>";
  document.getElementById("sortOut").innerHTML = out;
}

</script>

</body>
</html>
```

Output:


127.0.0.1:3000/lab-6/index.html
http://127.0.0.1:3000/lab-6/index.html
Exercise 1: Operators
2 + 3 = 5
"hello" + " bvm" = hello bvm
2 + "MB" = 2MB
3 - 2 = 1
3 - "MB" = NaN
3 * 2 = 6
5 / 2 = 2.5
2 / 0 = Infinity
5 % 2 = 1
2 % 0 = NaN
x = 1; x++ = 1
x now = 2
y = ++x; y=3 x=3
y = x++; y=3 x=4
x = 5; x-- = 5
x now = 4
y = --x; y=3 x=3
Exercise 2: Bitwise Operators
a = 5, b = 3
a & b (AND) = 1
a | b (OR) = 7
a ^ b (XOR) = 6
~a (NOT) = -6
a << 1 (Left Shift) = 10
a >> 1 (Right Shift) = 2
Exercise 3: Fibonacci Numbers
Enter N: 15 Show
First 15 Fibonacci numbers: 0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
Exercise 4: Prime Numbers in a Range
Start: 40 End: 140 Show
Primes between 40 and 140: 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131 137 139


127.0.0.1:3000/lab-6/index.html
http://127.0.0.1:3000/lab-6/index.html
x now = 4
y = --x; y=3 x=3
Exercise 2: Bitwise Operators
a = 5, b = 3
a & b (AND) = 1
a | b (OR) = 7
a ^ b (XOR) = 6
~a (NOT) = -6
a << 1 (Left Shift) = 10
a >> 1 (Right Shift) = 2
Exercise 3: Fibonacci Numbers
Enter N: 15 Show
First 15 Fibonacci numbers: 0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
Exercise 4: Prime Numbers in a Range
Start: 40 End: 140 Show
Primes between 40 and 140: 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131 137 139
Exercise 5: Array Methods
Enter numbers (comma separated): 7,5,6,9,8,4,1,2,3,5,9,0 Show
Original: 7,5,6,9,8,4,1,2,3,5,9,0
reverse(): 0,9,5,3,2,1,4,8,9,6,5,7
concat([10,20,30]): 7,5,6,9,8,4,1,2,3,5,9,0,10,20,30
sort() lexicographic: 0,1,2,3,4,5,5,6,7,8,9,9
sort() numeric: 0,1,2,3,4,5,5,6,7,8,9,9
join("-"): 7-5-6-9-8-4-1-2-3-5-9-0
Exercise 6: Sort Array
Enter numbers (comma separated): 3,6,9,5,5,4,1,4,11,2,5,7 Show
Original: 3,6,9,5,5,4,1,4,11,2,5,7
Ascending: 1,2,3,4,4,5,5,5,6,7,9,11
Descending: 11,9,7,6,5,5,5,4,4,3,2,1